

# UWE-4 Operational Strategy

Target Selection & Passes

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# Phase 1: The Selection Funnel

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To build a reliable detector, we need reliable data. We filter the entire amateur fleet through five critical layers to identify our “Golden Targets.”

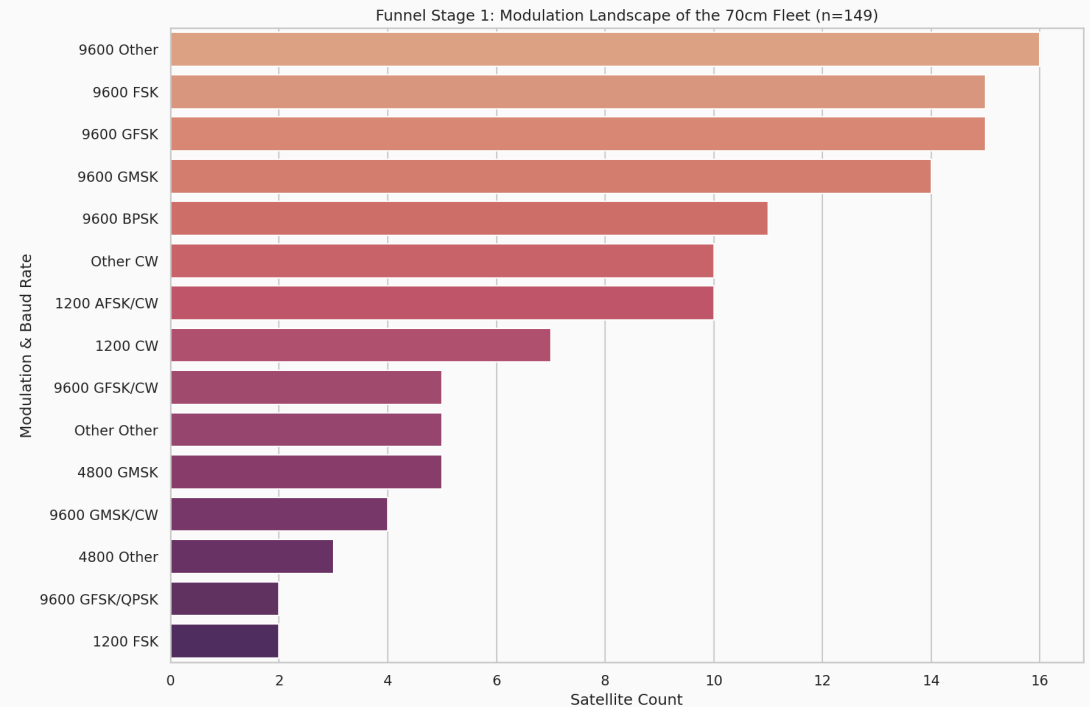
- **Status:** Must be confirmed ‘Alive’ in SatNOGS DB.
- **Band:** 433-438 MHz (70cm Amateur Band) for antenna compatibility.
- **Modulation:** High-rate 9600 bps FSK/GFSK for high-fidelity signal profiles.
- **Support:** Explicitly supported by `satnogs-decoders` (Kaitai) for automated parsing.
- **Visibility:** High-elevation passes ( $>30^\circ$ ) for clean, noise-free reception.

# Stage 1: The Fleet Landscape

We analyzed 149 active satellites in the target band to identify dominant communication standards.

## Observations:

- Huge variety in modulation types.
- Many legacy satellites use low-rate AFSK or CW.
- No single “universal” standard exists across the entire fleet.



## Stage 2: Technical Filtering

We narrow the funnel by applying technical constraints required for edge anomaly detection.

| Funnel Step   | Constraint                                 | Remaining |
|---------------|--------------------------------------------|-----------|
| 1. Bandwidth  | 70cm Amateur (433-438 MHz)                 | 149       |
| 2. Decodeable | Explicitly supported by `satnogs-decoders` | 7         |
| 3. High Rate  | 9600 bps GMSK/GFSK/FSK                     | 2         |

**Result:** We converged on a “Golden Cohort” of 2 targets that provide high-fidelity telemetry for ML training.

## Phase 2: Operational Planning

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# Heuristic: “Total Observable Time”

Pass count isn't enough. We rank satellites by the **total duration** they spend above 30° elevation over 48 hours.

## Satellite Scoring

**Goal:** Maximize data collection opportunity

**Logic:** Sum(Duration of all passes > 30°)

**Result:** Top Operational Targets

# Stage 3: Operational Convergence

Ranked by “Total Observable Time” over Beni Suef (>30° Elevation).

| Satellite    | Passes (48h) | Total Mins | Max El |
|--------------|--------------|------------|--------|
| INSPIRESat-1 | 4            | 7.8 m      | 85.8°  |
| UWE-4        | 2            | 5.8 m      | 87.4°  |

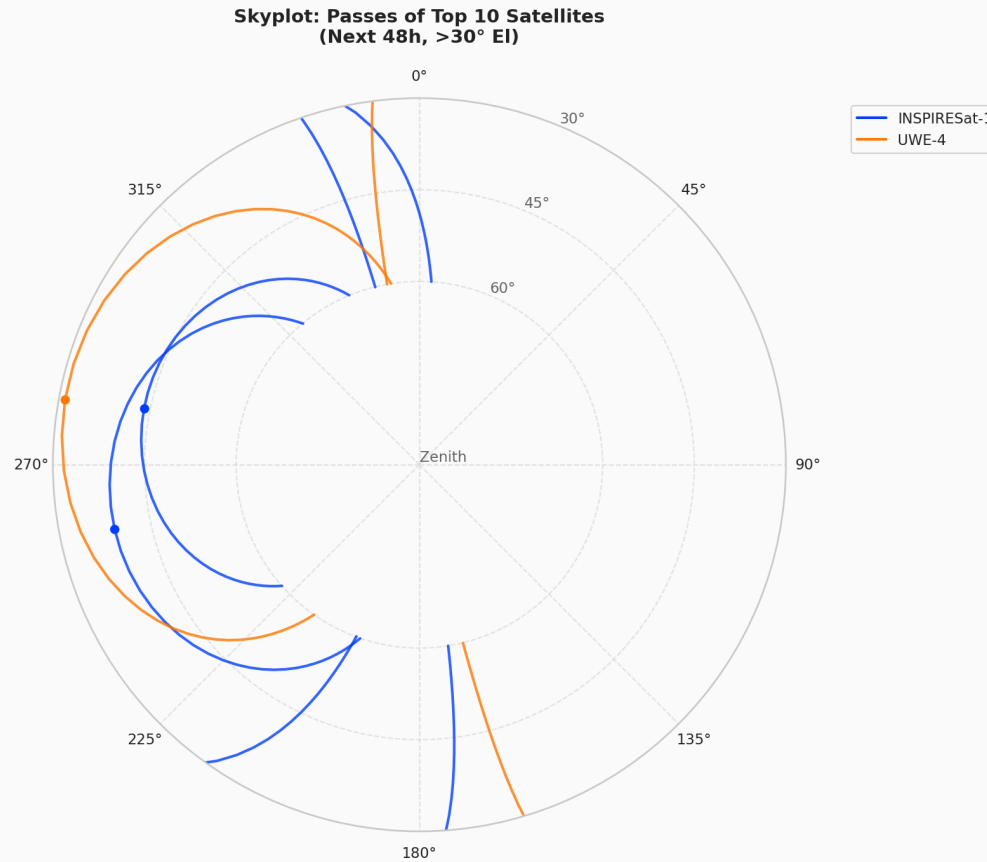
**Why UWE-4?** Despite having fewer passes than INSPIRESat-1, UWE-4 is our primary “Golden Path” due to its rich, multi-sensor telemetry (Panel temps, Battery currents, OBC health) which is essential for complex anomaly detection.

**\*Data Engineering Reality Check:** BugSat-1 was dropped despite high visibility due to undocumented protocol variations (US37 payload header) causing Kaitai parser failures. ML requires clean data; UWE-4 provides perfect compatibility and rich thermal/power telemetry.



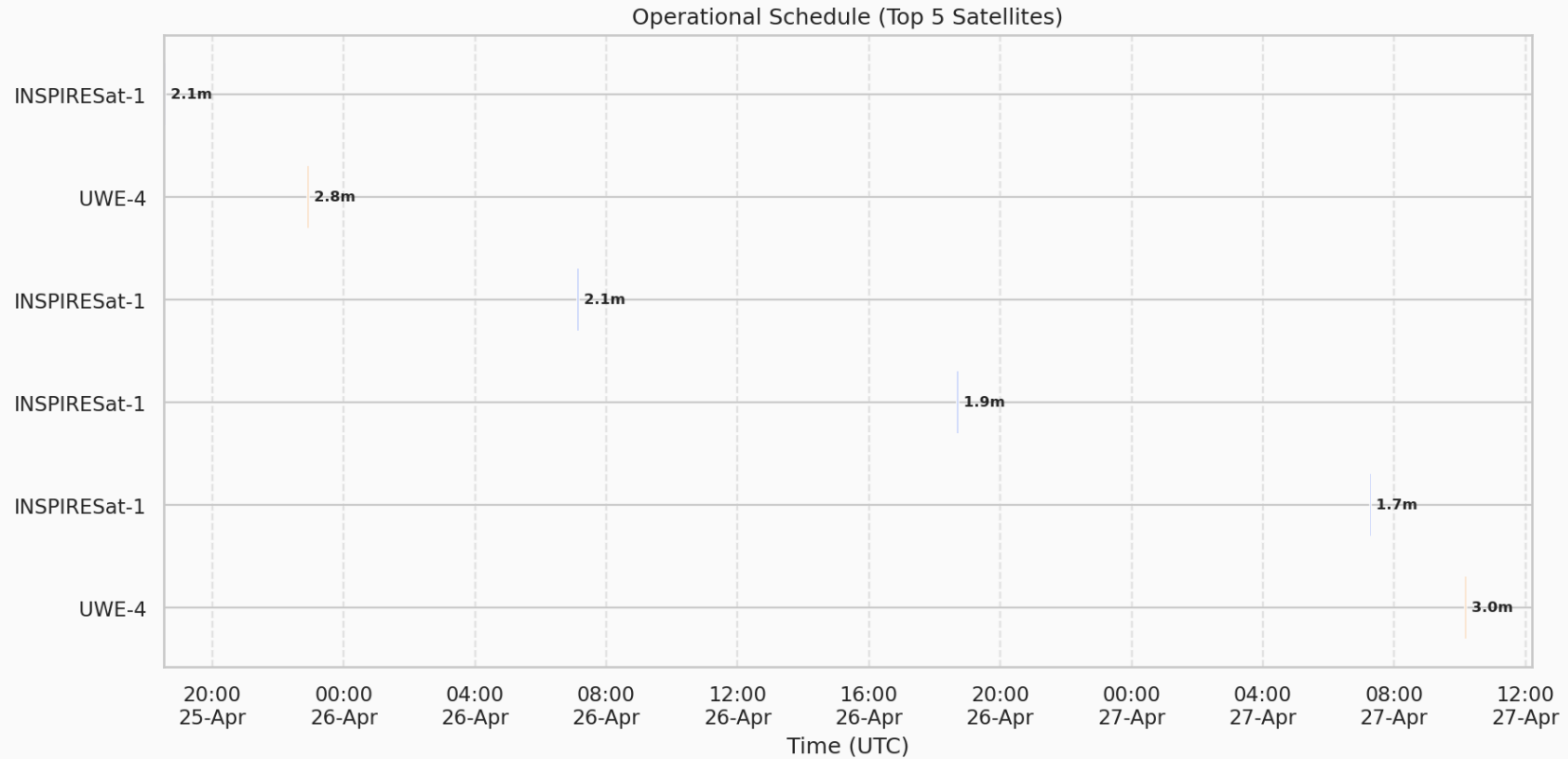
# Operational Reality: Skyplot

Where to point the antenna for the Top targets.



# Operational Reality: Schedule

When to operate the ground station (Next 48 Hours).



Two distinct environments sharing a single **Shared Core**.

## A. The Lab (Offline)

- Establish Baseline from history
- Source: SatNOGS DB
- **Action:** Train Autoencoder

## B. The Watchdog (Edge Deployment)

- Live anomaly detection (5 Docker Microservices)
- Source: Live Antenna -> SDR -> MQTT
- **Action:** FastAPI backend inference & SvelteKit dashboard

# Standardization: The Golden Features

*A universal interface for heterogeneous satellite hardware.*

| Feature           | Unit    | Description              |
|-------------------|---------|--------------------------|
| batt_voltage      | Volts   | Standardized from mV/ADC |
| batt_current      | Amps    | Charge/Discharge rate    |
| power_consumption | Watts   | Spacecraft power draw    |
| temp_obc          | Celsius | Main computer temp       |
| temp_batt_a/b     | Celsius | Battery thermal health   |
| temp_panel_z      | Celsius | Orbit-phase context      |
| uptime            | Seconds | Time since reset         |